

No late submission is allowed without a prior approval. 0 points from 10:40am.

- (13) Q1. Prove that any 2×2 game has a Nash equilibrium in mixed strategies.

You don't need to understand all the details, as some go beyond the scope of this course. The proof can be found on p121-122 of Chapter 6 in Tadelis (2013), for instance. So this question basically asks you to summarize that.

- (20) Q2. (Cournot Duopoly) Firms 1 and 2 produce a homogeneous product. Firm i chooses to produce quantity $q_i \in \mathbb{R}_+$ at a unit cost c_i , i.e., the total cost to produce quantity q_i is $c_i(q_i) = c_i q_i$. The market clearing price is $P(Q) = a - bQ$ where $Q = q_1 + q_2$ is the aggregated quantity.

- (5) (a) Write down firm i 's profit maximization problem and find best response function.
(5) (b) Find a Nash equilibrium.

For (c-d), assume $a = 150, b = 2, c_1 = c_2 = 30$.

- (5) (c) Calculate the social surplus (i.e., sum of payoffs of two firms) in equilibrium.
(5) (d) Now, a social planner wants to maximize social surplus. Find the aggregated quantity and social surplus for the socially efficient outcome.

- (20) Q3. (Stackelberg Duopoly) Two firms in Q2 have the same characteristics, but now firm 1 chooses q_1 first, and firm 2 observes q_1 and chooses q_2 .

- (15) (a) Find a subgame-perfect Nash equilibrium (Do not simplify constants, e.g., c_i .) and compare it with Q2-(b).
(5) (b) Now, for $a = 150, b = 2, c_1 = c_2 = 30$, calculate the social surplus.

- (17) Q4. (Hawk-Dove) The Hawk-Dove game is used to study conflict over a valuable resource. Two players simultaneously choose either Hawk (H) or Dove (D). The value of the resource is v . If both players choose H , they split the resource equally but each suffers an injury cost k , where $k > v/2$. If one player chooses H and the other chooses D , the Hawk player obtains the entire resource and the Dove player gets 0. If both players choose D , they split the resource equally.

- (2) (a) Write the payoff matrix in terms of v and k .
(5) (b) Now let $v = 10$ and $k = 6$. Find all pure-strategy Nash equilibria.
(10) (c) For the same parameter values, find all (nondegenerate) mixed-strategy Nash equilibria.

- (30) Q5. (Sequential-move Hawk-Dove) Continued from Q4, but now consider a sequential-move Hawk-Dove game where P1 moves first. Read the strategies from left to right.

- (5) (a) Draw a game tree.
(5) (b) What is the normal form of this game? Clearly show the payoff matrix.
(5) (c) Find all pure-strategy Nash equilibria.
(5) (d) Find all subgame-perfect Nash equilibria.
(5) (e) Is (H, dd) an SPNE? If not, is it an NE? Explain why.
(5) (f) Is (H, hd) an SPNE? If not, is it an NE? Explain why.